Design and Technology NEA

Aarav Thakur

10XO

Mr Laher

Problem 1

Delicate Plates

Since the majority of plates are made of glass, porcelain or ceramic, they make a highly unpleasant sound when scratched with a fork, they easily break and when they break, they shatter meaning that it is difficult to clean up all of the shards of plate and they shards are very dangerous as accidentally stepping on them could severely injure the person

A study by the NIH focusing on the injuries of minors has shown that 86% of lacerations are caused by minors stepped on shards of glass

One solution to this problem is tempered glass which is a form of glass which has been heat treated to become stronger than regular glass. One problem with this solution is that the plate can still break on impact making it not ideal for a plate

To solve this problem, I could make a plate which is hard, does not shatter and has an appropriate design and still maintain a premium feel.

Potential Client: My mother

Potential Target Market: Restaurants

Problem 2

Heavy Guitars

The average weight of a guitar is around 3.6 kg and is almost 1 metre long this makes it difficult for people including many of my guitarist friends to practice in different places. Although their passion for guitar pushes them to practice whenever they can, the heavy weight and large size of the guitar makes it difficult to do so. In addition, electric guitars require a large amount of gear to get the correct sound that one looks for and often times, this gear can be expensive and heavy further restricting people from practicing in places other than their home.

As shown in the table:
Throughout the majority of the current decade, guitar sales have been on the rise and haver increased by 6.4% in the past year alone

Total Guitar Sales

Year	Units	% Change	$ Retail	% Change	Avg. Price
2024	2,630,950	6.4%	$1,070,244,000	7.0%	$433
2023	2,472,700	-.3%	$1,000,676,130	11.4%	$403
2022	2,489,390	-.3%	$934,971,450	11.4%	$372
2021	2,496,185	5.0%	$839,000,000	2.2%	$353
2020	2,377,310	1.5%	$820,746,000	21.0%	$361
2019	2,341,551	20.5%	$903,261,000	-1.9%	$386
2018	1,942,625	11.4%	$921,057,000	.13%	$529
2017	1,742,498	5.6%	$922,280,000	-0.1%	$529
2016	1,648,595	23.3%	$923,522,000	21.2%	$560

A solution to this is problem to make a hollow guitar body which significantly reduce the weight of the guitar. However, hollow-body guitars have a different tone from a full body guitar.

Ways I could help:
When designing the guitar I must take into account everything such as but not limited to tone, feel and comfort while still optimising its smaller size and lightweight.

Potential Clients: Shiv, Ubai, Kenda, Rocco

Potential Target Market: Guitarists

Problem 3

Robotics Education

Although movement such as FIRST have revolutionized robotics among high schoolers, organising events and competitions all around the world encouraging more teenagers to engage in robotics, it has a long way to go before reaching all corners of the earth (which is a cube). For example, in countries like portugal, spain and belarus, there are no initiatives which encourage robotics. This has become hindrance to many people who are actually interested in the subject but it is either too expensive or too distant to be worthwhile

- By 2030, there could be a **global shortage of over 85 million skilled workers**, many in STEM fields. **Source:** Korn Ferry – *The Global Talent Crunch report*
- **Only about 40% of schools worldwide** have access to adequate STEM learning resources.**Source:** UNESCO – global STEM education reports
- commercial robotics kits cost **$200–$500 per student and hundreds of thousands for competitive robotics. Source:** market data from companies like LEGO (e.g., LEGO Mindstorms) and VEX Robotics kits.
- Around **2.6 billion people globally still lack reliable internet access**. **Source:** International Telecommunication Union

Ways I could help:

I could create a cheap robotics kit which can be easily distributed through rural regions such as villages in india and teach them about engineering and robotics.

Potential Client: Faris Tarraf

Potential Target Market: Robotics enthusiasts and young child

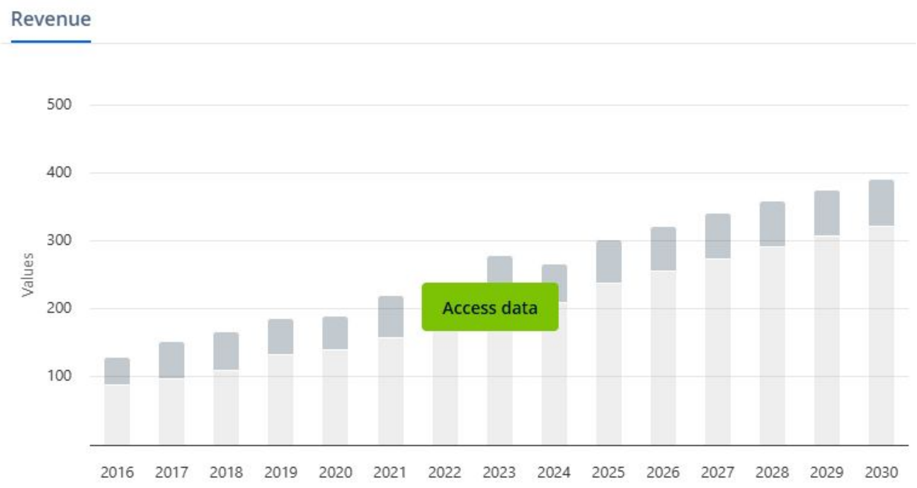
In-depth research

Benefits of robotics education:

Problems with robotics education

Target User data

Classroom limitations



Existing products:

Conclusison

Design brief

I will be designing a product that satisfies the needs and demands of middle and high school robotics enthusiasts. This product will be used over a period of 2 weeks to 2 months dispensing on how much time they spend on the kit so it must meet the requirements for that. The costumer range will be between the ages of 12-18 as often they want to get started and dive deep into robotics but can only rely on small allowance money to do so.

In addition, I would like my product to be aesthetically pleasing as possible, to a meet this goal I will make the product have a clean and well-managed design, making it have a sleek appeal. The finish will be matte to give it a bolder appearance and to remove any extra stickiness that glossy colour may incur. Since tennagers often are developing their personalities, I will keep the designs simplistic..

Furthermore, the product will need to meet a range of requirements and specific needs for my client. Therefore I will need to research into my clients needs and wants through an in-depth interview and questionnaire and also into the wider publics needs and wants through a simple, structured questionnaire.

The product must also be ergonomically designed and has the potential to suit multiple users, including my client and potential sellers. To do this I will do my own research on existing products, analysing the advantages and disadvantages of each product.

This products main use will be to educate teenagers on the principles of robotics for a low price so the need for it to be functional, minimalistic and cost efficient is necessary.Hence, the product will be made of reliable and cheap materials such as PLA and plywood.

The product will also be manufactured using machinery only available within the department. Therefore any manufacturing process and materials must be used efficiently so that it minimises cost and therefore maximises profit from the sales. The materials must also be available in the department meaning that every item must be used to its maximum potential to achieve the goal of making this product.

Research plan

What I plan to research	How I plan to gather the information	Why I will research it How will it help me with my project
Client information	Primary research: Interview	It will help me understand what my client wants
Aesthetics	Primary research: Interview Secondary research: Internet	It will help me design my product in an appealing way
Cost	Primary research: Interview Secondary research: Internet	It will help me set a target price on the product
Environment	Secondary research: Internet	It will help me prevent harm to the Environment
Ergonomics and anthropometric	Primary research: Client measurements	It will help me make my product comfortable to hold
Existing products	Secondary research: Internet	It will help me get ideas for my product
Size	Primary research: measurements Secondary research: Internet	It will help me set constraints on my design
Safety	Secondary research: Mr Laher and internet	It will help me prevent injury when interacting with my product
Function	Primary research: Interview Secondary research: Internet	It will help me understand what features my guitar should have
Materials and finishes	Primary research: Experimentation	It will help me decide which materials to use for my product
Manufacturing processes	Primary research: Experimentation	It will help me plan how I will build my product

Have you done any robotics or electronics before?
Yes I have.

What was your experience like?
It was very enjoyable

What made you interested in robotics in the first place?
I first got into Robotics when I was young through a Lego Mindstorms Course my parents signed me up for.

What specifically about being in Portugal has made it hard to access robotics?
The lack of proper Robotics Teams near me, as well as the language barrier.

Have you ever tried to build something electronic or mechanical on your own?
Yes. Mechanical (I believe it would count as mechanical). A Pinewood Derby Car.

How much would you (or your parents) realistically be willing to spend on a robotics kit?
I would say 50 Euros, give or take.

Would you prefer to follow step-by-step instructions, or figure things out yourself with hints?
I'd prefer clear instructions, instructions for individual mechanisms/features, and the ability to just figure things out myself.

Do you prefer a clean, polished look or a more barebones, mechanical look?
Both are okay, but a clean look is nicer.

Will you be using this alone or with an adult nearby?
I will be using it alone

Do you have any allergies or sensitivities to materials like certain plastics or strong-smelling paints?
Not that I know of.

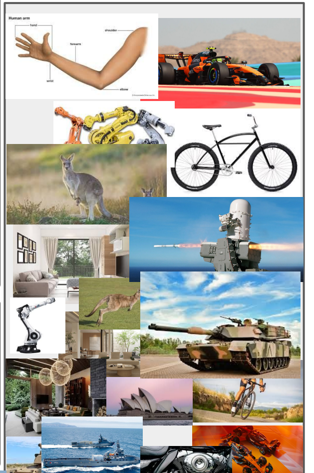
Name: Faris Tarraf
Gender Male
Age: 13
Location: Braga, Portugal
Language spoken : English, Arabic
Date of birth: June 11th 2012
Handedness: Right

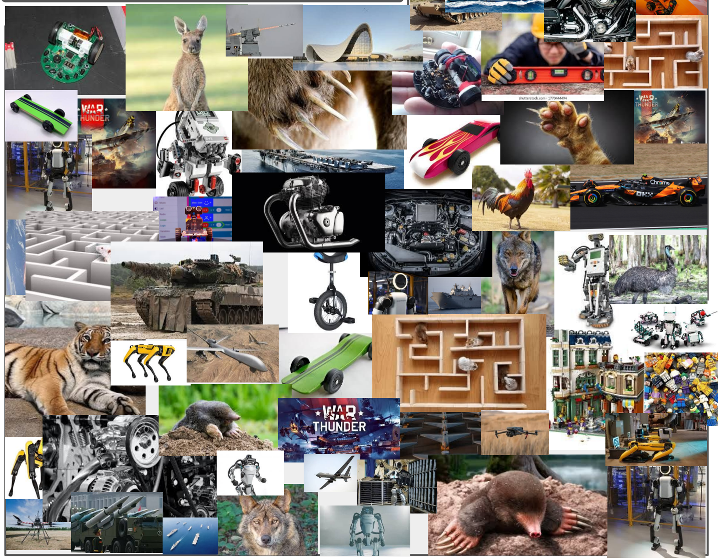
His hobbies include gaming and coding. However, due to his current relocation to portugal, he has been unable to participate in robotics. He often complains about expensive kits and the lack of programs within his area. For this reason I am making a cheap robotics kit which can be used to learn about basic engineering concepts

Client Needs and Wants:

Needs:
circuit stuff, motors, and building materials (like stuff to build the chassis), cheap and fun
Wants:
External sensors such as magnetometers cameras

Target audience
My target audience is children and teenagers who are interested in engineering or robotics but cannot afford to do so practically.





Aesthetics

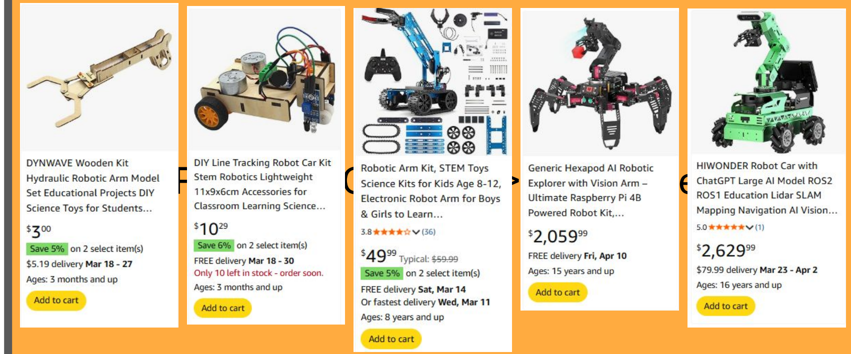
The robotics kit on the right uses blue parts which research has shown boosts creativity. This aids in the user's engagement and boost conceptualisation. Additionally, the continuous treads and the metallic coating makes the robot feel more like a tank, further interesting children and teenagers,

The use of the ultrasound sensors as eyes makes the robot appear more human forming a stronger connection between the user and the robot. Furthermore, .

Colour Psychology: As referenced in the first aesthetic analysis of a product, certain colours abet the user to express certain emotions. For example, silver (which most metals are coloured) induces a feeling of sophistication.Hence, I will use blue as a main colour and red as accents;

Biomimicry: Biomimicry is the use of the natural world to solve problems that we ourselves have. Incorporating the natural world into my product, it will teach the user that a great source of inspiration for solving real world problems is to look at nature as it often has already solved the same problem but in a different context. For example, the bullet train often made large booms whenever passing through tunnel. So engineers took to the kingfisher bird for inspiration which ultimately helped with the final design We can apply the same concept to this kit. The "animal" that most people interact with are human so making a humanoid robot would build a greater connection between the robot it the user.

Cost



Cost of my product: The cost of the product will be determined materials I use, the current price of similar products on the market, and the time and labour that will go into making the product. Due to one of my product targets is being cheap, I will have to price the kit in the middle to low price range.

Environment and sustainability

Fossil fuels are fuels which were formed over millions of years from dead animals and plants. Thes often use combustion to release energy. This releases toxic gases such as sulphur dioxide and carbon dioxide into the atmosphere and contribute to global warming. Nuclear energy is energy derived from the nuclear decay of enriched uranium 235. This produces radioactive waste which is difficult to dispose of and takes centuries to decompose. Renewable energy is energy derived from renewable sources such as wind and the sun but is unpredictable and only produces energy in intervals. A finite resource is a resource which will eventually run out. An infinite resource is a resource which is able to be renews at a rate in which the new resource can replace the old resource. Though this does not exist, we should preserve the energy we do have. For this reason, I have decided to an use easy to manufacture material such as PLA which does not require much energy to deform but remains still at room temperature.

The recycling symbol three arrows symbolize the process of collecting, manufacturing, and purchasing products made from recycled materials. Recycling is needed to conserve natural resources like trees and minerals by using materials which were previously used to make products; reduce energy consumption and greenhouse gas emissions, decrease pollution from landfills and waste, protect ecosystems and wildlife.

LCA is a method to determine a product's impact on the environment. LCA looks at the entire life cycle of the product to prevent one impact from being shifted to another. The LCA is a standardise industrial practice which is governed by ISO 14040 and 14044.

What are the 6Rs:

1) Rethink (think about using non-recyclable materials before using them)
2) Refuse: try your best to avoid using non-recyclable materials
3) Reduce: limit the amount of non-recyclable
4) Reuse: limit the amount of materials you waste by using scrap material which otherwise would have been considered as trash
5) Repair: try fixing things before throwing them out
6) Recycle: if it is not possible to reuse the materials, try recycling the materials

Materials that can be recycled:

1) Paper
2) Cardboard
3) Wood
4) Metal

Materials which a cannot be recycled:

1) Plastic
2) Styrofoam
3) Ceramic

Product Analysis

Aesthetics:
The robotis mainly black with yellow and blue accents. This along with its barebones nature makes the product look aesthetically pleasing while remaining functional

Size:
The robot is a compact robot car for a tabletop. Hence can be used on desks and is easy to store and transport.

DR CLINT
Wonderful product as a STEM kit!!!
Reviewed in the United States on February 6, 2026
Verified Purchase
This Smart Robot was my introduction to Elegoo products and the experience has made me a diehard fan of Elegoo. I bought this for my 9 year old son as a gift for the holidays. This was my attempt to give him gifts would peak his interest enough to get him away from all the games and electric devices and this did the trick. At first when I first opened it I wondered if it was to much for him to assemble but boy was I worng. He enjoyed every moment of it and has not stopped using it. Its so much fun that everyone in the house is enjoying it. It was designed and packaged so well and the instuctions were very easy to follow. Love the care they took and the packaging was neat and well organized. In its default state its packed with features but the intro to Arduino IDE increases its value immensely making it possible to add more components, features and functionality. This has already paid for itself times over and its value for money is undeniable!!

Happy Shopper
so-so
Reviewed in the United States on December 28, 2025
Verified Purchase
The good: In spite of it seeming like a toy, it is a very complex electro-mechanical device. The kit is assembled with well illustrated instructions.
The so so: I don't think the average 8-12 year old could assemble without damaging or mis wiring it because of the tiny parts and no knowledge of what the component boards do. My 11 year old granddaughter who loves science, math and stem looked at it and walked away. I built it and then she took over running, programming and connecting to her phone. Then......
The bad: It started falling apart. The camera module fell off, the drive motors came loose because the fasteners all unscrewed from the turning and jerking and vibration. The smallest of nuts were lost forever. My assembly issue was to overtighten and risk breaking boards or not tight enough and have them come loose.
In conclusion: Its relatively inexpensive and does work. I think I would use Locktite on screws. Also wrenches to hold tiny nuts should be included.

Safety:
Low voltage electronics (Arduino boards and small motors). Requires supervision for younger users due to small components and wiring. It is unlikely that someones finger get stuck in any places

Cost:
The cost of the product is around 100$. However, this is only the cost within the US. I believe this is a fair price as it offers high quality parts at a feasible price point.

Environment:
The product is mostly acrylic and electronics. Hence, the product itself is not environmentally safe. However, the parts could be reused in separate projects making the product less environmentally threatening. However, the acrylic will have to be thrown away once used as it cannot be used for be used for any other project.

Function:
Allows students to **build and program robots using Arduino**.
Often includes sensors such as ultrasonic distance sensors and line followers.
Teaches coding, electronics, and robotics principles.

Customer:
The product has clearly been designed for a customer to build themselves as it include not only the parts that are needed but also the physical tools that would be used

Product analysis

Size:
Approximately 18.3 × 4.7 × 11.2 inches.

Desktop-sized robotic arm suitable for laboratory or classroom work.

Cost:
Mid to high price for an educational robotics kit (around $150–$200).This high of a cost is due to multiple servo motors and metal construction. I believe this is unreasonable as it provides little educational value compared ot other competitors.

Safety:
Moving points such as joints and grippers are possible spaces for children to get their fingers stuck so safety precaution such as turning of the orbot must be take. However, th low voltage prevents this from causing real harm

Customer:
Targeted at older students, robotics hobbyists, and educators. Recommended age 14+ due to complexity and is used for learning robotics, programming, and automation.

Aesthetic:
Industrial robotic arm design made from metal. The used of CNC'ed metal gives the robot a professional appearance similar to small factory robots. Exposed mechanical joints emphasize engineering structure.

Environment:
Durable aluminum construction increases product lifespan. However, electronic components require proper disposal when damaged. Since the product can be used for a long time, it reduces the waste cause by the product.

Function:
The robot is a 6-degree-of-freedom robotic arm which can be controlled via PC, mobile app, or wireless controller. Used for pick-and-place tasks and robotics programming practice.

Tom Carroll

★★★★★ **Quality Construction. Clear Assembly Videos**

Reviewed in the United States on March 2, 2018

Edition: LeArm | Verified Purchase

When I received the LeArm kit from Amazon, I was amazed at the quality of the arm, from the heavy metal base, industrial quality ball-bearing, good servos and the very clear assembly instruction videos on the Internet. It assembles easily in a few hours. I would recommend this arm over any other robot arm for robot experimenters as it will last. I intended to write about it in my monthly column in a robotics magazine and was waiting for their version of the 5-DOF hands, another manufacturer's version of which is shown in the photo. I never received those hands from LeWanSoul. I did receive their X-Arm shown to the right in the photo and found it to also be of excellent quality.

However, after a year of experimentation and demonstrations for others, I have found both of these robot arms far higher quality than any of the other robot arm kits is have tested. The LeArm uses separate 'standard' servos, each of which must be connected to a controller or Raspberry Pi or Arduino board. Depending on the position of the arm, I could manipulate almost a pound. Programming of the LeArm is fairly simple as you are dealing with standard servos that require basic pulse-width signal inputs.

The X-Arm has very similar construction but you can look at both robot arms at the Amazon site and see the physical similarities and differences in the two arms. Both use the same quality ball bearing base, though the metal structure is a bit different and the gripper/claw is identical to the other. This arm uses their LX-15D intelligent serial bus servos that can be 'daisy-chained' and the final end of the three wires of the daisy chain transfers the servo's address (each servo has a number of 1 to 6) and power through the three wires. (You can see the small connector on the end of the white cable in my photo) Your microcontroller addresses each servo independently through that cable, very similar to the Robotis Dynamixel servos. The higher cost is due to the cost of the high-cost intelligent servos.

I had a bit of trouble programming the arm with a RaspberryPi but soon got the hang of it. I ended up writing a bit of code to use it with Python, directly.

The only issue I might say is using the tiny metric screws that were all in one bag. I found some that were too long so I used washers in mounting them to the small aluminum cylinders between the servos. One of the threaded holes in the long cylinder was a few degrees off 90 degrees so I just left the screw off. The video instructions were fine though I would have preferred a written manual. I found myself 'rewinding' the video to listen again 'what the girl really said.' I also would have liked to test their two hand models as theirs seems so much better made than the other company's hands shown in my photo.

I also have their 6, 16, 24 and 32 channel servo controllers that have extensively come into use with my demonstrations. The LSC-6 controller is furnished with the LeArm and is used with the hand controller (they call it a "handle") to control the arm wirelessly. Note the receiver board in the photo of their photo of the arm.

I would highly recommend the LewanSoul robotic arms and controllers and servo testers for students from high school to advanced university instruction, as well as home experimenters. Either of their servo models are well made with metal gears and have yet to fail for me. Their web site offers much material on their products, and they have been around long enough for one to find all sorts of articles about uses of the LewanSoul robot arms and controllers.

^ Read less

blakebart

★★★★★ **Extremely well designed and manufactured 6DoF robotic arm with remote Bluetooth to iPhone/Android connectivity**

Reviewed in the United States on June 26, 2018

Edition: LeArm | Verified Purchase

I'm surprised more people haven't reviewed this robotic arm kit. I found the kit extremely well thought out. The parts machining and anodizing are just beautiful. A very important aspect of this design is the turntable that the arm sits on. It is built using a large diameter industrial quality bearing which provides a very solid base for the arm to operate on and provide servo-controlled rotation. The servos are top quality, very smooth operation and very powerful. The gripper is pre-assembled, a nice touch, and the joint design is quite clever. LewanSoul provides excellent tutorial videos, and an extremely helpful assembly video using 3D modeling to show how things go together in sequence. Another reviewer noted the importance of sorting out all the fastener hardware, that really helped smooth the build for me. The Bluetooth connection to the free iPhone app worked great, another impressive feature of this kit, especially for the money. The PS2 controller worked fine, though I prefer the iPhone app. There is also PC interface software that is provided although I have not tried that. I'm particularly interested in connecting an Arduino to the arm and writing sketches for controlling LeArm, and Aaron at LewanSoul showed me where to find the necessary information for that. LewanSoul has been very responsive answering my questions. Oh, and check out the video for LeHand on the LewanSoul website.

46 people found this helpful

Product Analysis

Aesthetics:
Modern educational robot design with colored metal parts and wheels. Friendly appearance suitable for classrooms and younger students. Includes a display and sensors that give the robot an interactive look.

SDMF1919
Very frustrating at first
Reviewed in the United States on December 27, 2024
Size: mBot Coding Box | Verified Purchase
It was easy to build. My 10 year old did it by himself. It took us about 4 hours to figure out how to control it though. The app in the android store is only compatible with really old phones and then once you find an old phone and get it installed, it doesn't work anyway. The chrome browser app doesn't work either. Use the QR code on the mblocks website to download and install the updated android app. Then it will finally work on your current device.

Aleiree
Perfect for 11 year old
Reviewed in the United States on January 11, 2026
Size: mBot Blue | Verified Purchase
This was an easy and fun build. My kid has easily gotten it to do things, but probably because he uses similar apps/code at school. He really enjoys it. The only downside is that it doesn't always reconnect via Bluetooth on its own and he needs a little help with that. It's perfect for his age (11) and I'd buy again for my other kids.

Customer:
Designed for STEM education, schools, and coding programs and is suitable for learners aged around 12+. It is often used in robotics clubs and coding classes.

Environment:
Although it uses recyclable aluminium, it also uses ABS which can harm the environment. Since the kit uses rechargeable batteries and modular parts, nothing has to be thrown aways as it can easily be recharged, repaired or upgraded.

Cost:
Higher-end educational robotics kit (around $150–$250).

Higher price reflects advanced sensors, controller board, and software ecosystem.

Size:
Compact mobile robot about **90 × 180 × 130 mm** makes it well suited for classroom desks and small testing environments.

Safety:
Rounded edges and enclosed electronics make it safer for students as it prevents them from getting cuts or from getting shocked.

Materials:
Aluminum structural parts
ABS plastic casing
Electronic sensors and PCB control boards
DC motors and wheels.

Feature:
The kit consists of a simple 2 wheel drive which can easily be programmed in either scratch or C++. This allows the user to not only control the robot remotely but also autonomously

Scroll Saw:
The scroll saw has a sharp edge which quickly oscillates up and down to cut through wood

Advantages	Disadvantage
• Allows for curved and intricate pattern	• Not as versatile as other saws • Can only cut a single angle

Solder:
The Soldering iron has a hot tip which is used to melt solder and solidify the connection between a wire and the pins

Advantages	Disadvantage
• Can be done in under 30 seconds • The joints can withstand forces	• Creates fumes which contain lead (Depending on the solder)

Vacuum Forming:
In vacuum forming, the former heats a plastic sheet until it is soft and flexible. It then uses a vacuum to suck the plastic around a mold.

Advantages	Disadvantage
• Quick to make pieces • Simple to use	• Only useful with thermoplastics • Cannot create intricate shapes

Heat Press:
The heat press heats the textile and presses a special material onto it, forming a design on the surface.

Advantages	Disadvantage
• Easy to use • Quick to print designs • Low-Cost	• Could lead to inconsistent results

Belt Sander:
A long piece of safing paper spins around a drum, shaving off chips on the surface of the wood.

Advantages	Disadvantage
• Quick to sand parts down • Simple to use	• The surface still remains slightly rough • Cannot reach all corners of a wooden piece

Strip heater:
The wire heats up allowing it to easily cut soft materials such as styrofoam.

Advantages	Disadvantage
• Quick to make models • Good for prototypes	• Not accurate • Only works for certain materials

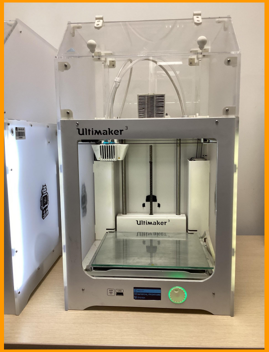
Pillar Drill:
The drill bit at the top rotates very quickly to drill through a material. The lever on the right pulls the drill straight down into the part.

Advantages	Disadvantage
• Quick • Easy to drill	• Can only drill vertically • Can be unsafe in the part is not clamped

3D Printer:

The nozzle on the top heats a plastic until it become s malleable as lays the material down onto the print bed

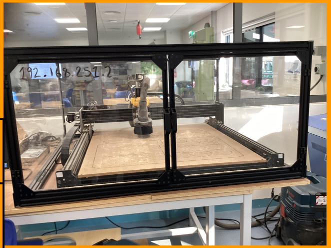
Advantages	Disadvantage
• Can make intricate pieces • Good for prototypes	• Only can make them in plastics • Slow • The print can fail



Laser Cutter:
The laser cutter burns through the wood to cuts certain pieces out of it by using a high power laser.

Advantages	Disadvantage
• Quick to make pieces • Simple to use	• Only works in 2 dimensions • Burns the material

CNC machine:
The drill bit at the top rotates very quickly. The Drill bit is moved around to remove certain parts of the material



Advantages	Disadvantage
• Good for intricate designs • Simple to use	• Slow • Only works in 2D • Only works for certain materials

Hand Tools:
The hand tools allow us to create designs using physical tools such as saws and handheld drills which gives us the most amount of control.

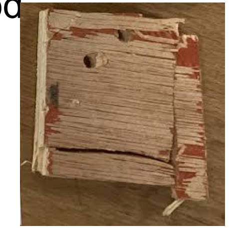
Advantages	Disadvantage
• Uses Full dexterity	• Takes time to make • Difficult to make it

Vinyl Cutter:
The machine quickly cuts the vinyl using a very sharp blade and moves the vinyl by moving the material vertically and the blade horizontally

Advantages	Disadvantage
• Can make intricate pieces • Good for prototypes	• Only can make them in vinyl • Only works on a relatively small scale

Material	Advantages	Disadvantages	Uses
Plywood	• Strong and durable • Resistant to warping • Lightweight • Cheap	• Susceptible to delamination • Hard to make intricate cuts • Inconsistent quality • Sometimes contain gaps	• Roof decking • Cabinets • Wall paneling
Oak	• Dense • Aesthetically pleasing grain grain • Resistant to wear • Long Lifespan	• Expensive • Heavy • Can Split • Can wear tools	• Furniture • Floors • Doors • Veneers
Pine	• Cheap • Easy to cut • Light • Aesthetically pleasing grain	• Soft • Can warp • Not moisture resistant	• Furniture • Shelves • Framing • Paneling
MDF	• Smooth • No grain • Cheap • Easy to cut	• Heavy • Poor moisture resistance • Lots of dust	• Cabinet • Mouldings • Speakers • Panels
Acrylic	• Clear • Light • Available in different colours • Weather resistant	• Scratches • Hard to machine • Brittle	• Signs • Screens • Lightboxes • Aquariums
PLA	• Useful for 3D printing • Biodegradable • Available in many colours	• Brittle • Not useful long term • Low strength	• Prototypes • Model Parts • Replacement Parts
Aluminium	• Lightweight metal • Resistant to corrosion • Easy to machine • Recyclable	• Soft • Expensive • Lower Strength	• Robot frames • Extrusions • Screws • Heat Sinks
Steel	• Strong • Cheap • Easy to machine • Long lasting	• Heavy • Rusts • Conductor • Hard to cut	• Cars • Appliances • Beams • Tools

Material Testing

Material Name	Scratch Test	Dent Test	Drilling Test	Cutting test
Acrylic 	• Does Not scratch with a fingernail • Captured Fingerprints • Scratches with sharp objects • Scratches are only visible when shone against a light	• Brittle • Easily Dents • Quickly fractures once dented	• Quite resistant to drilling • Large amount of residue builds up around the drill hole	• Soft • Heats up quite a bit • Results in a coarse cut
MDF 	• Does not scratch	• Difficult to dent	• Easy to drill	• Quite soft • Cuts were clean
Plywood 	• Brittle • Breaks along the grain • Easily Deforms • Scratches across the grain	• Flakes • Dents Easily	• Easy to drill	• Cuts were not clean • Easy to cut

Finger Joint Cube

Making a cube with finger joints by hand was a tedious and repetitive process which did not result in a particularly good product. First, I made borders on each wood piece using the side of another wood piece. Then I marked which parts I would need to cut. 2 sides would need the all the middler sections cut out, 2 sides for need all the edge sections cut out and and the other 2 sides would need half and half cut out. To cut out the sections, I used a coping saw for coarse cuts and sandpaper for finer cuts.With the coping saw, I often cut on or beyond the marking leading to inconsistent pins and slots. As a result, I had to test each joint as many of them did not fit on the first test. Once all of the joints fit, many holes were left between ach joint which I patched up and fixed in place using wood glue. Finally, I sanded everything down to get smooth surfaces and rounded edges.

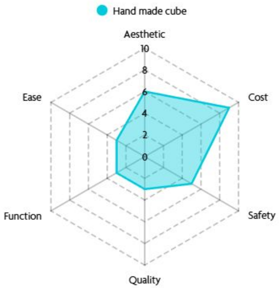
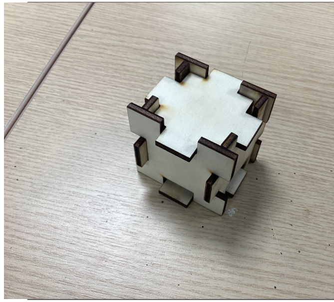
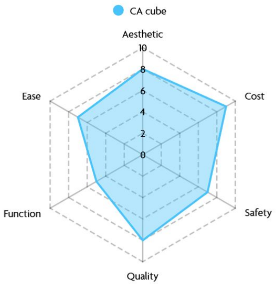


Photo of wooden box

Making a cube with finger joints with CAD and CAM was drastically easier than doing it by hand. To design the parts needed for the cube, I used Techsoft. Techsoft intuitive design made the process of marking the parts a whole lot easier. Once the CAD was made, I uploaded it to a USB stick as a DXF file where I plugged it into a separate computer where, using the values written on a notebook, I calibrated the power of the laser cutter. I then uploaded my design and turned on the laser cutter. Due to the age of the school's laser cutter, I had to move the laser head out of the corner before using it. Once the cut was complete, I literally added some glue on the edges of each piece that would come in contact with another part. Then I sanded everything down ti get smooth surfaces and rounded edges





Size

The robot kit must be able to fit into a small box of around 300mm x 200mm 50mm with the final robot being able to easily fit on a small desk. The individual parts must be able to fix in the user's hand for easier manipulation of elements. Hence the final bot will be around 200mm x 200mm x 200mm.

Function

Problem research::

The kit must be able to accept some sort of input from a single or multiple devices and output a resulting motion or action. After completing the kit, the user must understand both basic and relatively advanced concepts of robotics

Components
A cheap microprocessor will be used as the brains of the whole robot as it is cheap and readily available. Alongside that, I will use gold MG946R servos because of their all metal gear which prevent gear skipping along with a TT motor which permits ample speed without being dangerous.

Skills it will teach: The kit will teach the use about basic concepts such as gears, belts ropes, with motors, scissor lifts along with programming skills such as sensor control, PIDF loops. These allow the user to bit only build their mechanical skills but also their coding skills

Safety

Safety Standards my product will have to pass:
CPSIA
CE Marking
BSI EN 60204-1
IEC 62368-1

The BSI (British Standards Institution) is the national body which regulates technical and Safety standards across the country. It also is the body which test products for their safety and reliability and certifies them as legally sellable. They do regular tests on each product to ensure that the product still remains reliable and safe. When designing my product, I will make sure that it follows the BSI standard such as protection against electric shocks by using suitable insulation, fire hazards by using safe voltages, mechanical risks and hazards from environmental conditions use. From this, my product will receive the BSI Kitemark and this will ensure the consumer that the product is safe to use and marketable in england

The CPSIA (Consumer Product Safety Improvement Act) ensure that the product does not include any or limited harmful chemicals such as lead and Phthalates to protect children. It also mandates that the product has to go through a form of third party testing to ensure it does not harm the child.

The CE mark on products states that the product is allowed to be sold in the EU (European Union) and EEA (European Economic Area). For a product to be legally sold in the EU and the UK it must apass the EU's safety, health and environmental standards. My product must always abide by these standards as to permit sale within europe which (as stated in my in-depth research) is an extremely large market.

To keep my product safe in the hands of children I will employ a number of safety features which can be separate into 2 categories: electrical and mechanical. To prevent the children from getting shocked, I will use a battery with less than 12V, Fuses to prevent overcurrents, use proper insulation on wires, clearly label the power input and place the batteries in a safe holder. Mechanically, I will fillet the edges, make sure that the screws cannot fall out by using either threaded inserts or nuts to secure the parts, use large parts to prevent young children from choking and use safe and non toxic materials.such as PLA

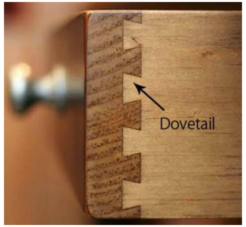
Name	Description	Application	Advantage	Disadvantage
Spray Paint 	Spray painting is a type of painting where the paint is atomised before being released into the air and landing on the part	• Shake the spray can • Start spraying the paint while slowly moving over the part from side to side	• Fast • Smooth finish • Reaches difficult to reach places • Less physical effort: • Versatility:	• Wastes a large amount of paint • Requires preparation • Not safe • Expensive
Varnish 	A clear, protective coating made of resin, solvent, and a drying oil that provides a hard, lustrous finish to surfaces like wood	• Dip the brush into the white varnish • Varnish the part by brushing along the grain • Wait 5 min before doing another coat	• Protection • Durability • Appearance • Ease of Application	• Slow Drying • Yellowing • Difficult Cleanup • Bubbles could form • Multiple coats • Strong Odour
Wax 	Wax is a a protective coating made from natural or synthetic waxes that enhances the wood's natural grain, and offers moisture resistance	• Use a piece of cloth to rub the wax • Rub the part with the cloth	• Aesthetic • Easy To Apply • Environmentally Friendly	• Weak • Requires maintenance • Difficult to remove • Wears off quickly
Stain 	Stain is a coloured substance which ‘bleeds’ into wood, changing its colour while still showing the grain	• Dip the brush into the stain • Stain the part by brushing along the grain • Wait 5 min before doing another coat	• Aesthetic • Protection against moisture, mildew and sun • Easy to apply • Cheap	• Has to be replaced • Does Not hide imperfections • Few colours •
Paint 	Paint is a coloured substance which form a decorative or protective layer over a patt	• Dip the brush into the paint • Paint the part by brushing along the grain	• Aesthetic • Durable • Easy to clean • Versitile • Eco-friendly • Cheap	• Difficult to remove • Requiress prep
Lacquer	Lacquer is a clear protective layer which provides a glossy finish	• Spray the part vertically	• Fast Drying • Burable • Aethetic	• Toxic • Yellowing

Fixing and construction methods

Wood

Finger Joint:
A finger joint cuts the wood in interlocking wedges, increasing surface area to create a stronger adhesive bond, spread out stresses and is versatile

Dovetail joint:
Instead of cutting straight wedges, flared tails' are cut out of the wood which intersect to create a strong joint. This design makes them less susceptible to lateral forces, they are more aesthetic than the finger joint and are more durable.

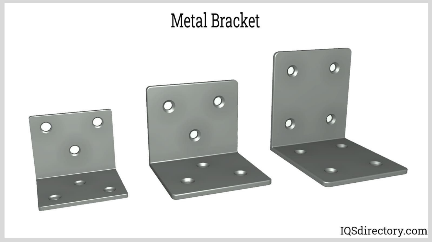


Butt joint,
a butt joint is the simplest type of wood joint making it quick to do but makes the joint weak and it may be difficult to perfectly align

Adhesive:
An adhesive is a substance which connects 2 materials together. This allows 2 different materials to be joined together differently and distributing stress and are easy to apply

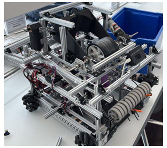
Metal

Metal Bracket:
Metal brackets are essential structural fasteners used to connect, support, and reinforce materials, commonly made from steel, aluminum, or iron. There are many types of metal bracket: L, Z, T, Flat.



Welding:
Welding works by melting base metals to physically bind the 2 pieces of metal together. This results in an extremely strong joint. However, the equipment required for this process is expensive

Rivet:
Rivets are pieces of metal which pass through holes within the metal. The end of the rivet is then deformed ,often by a hammer, creating a second head on the other side of the metal pieces. This creates a strong bond between the pieces of metal

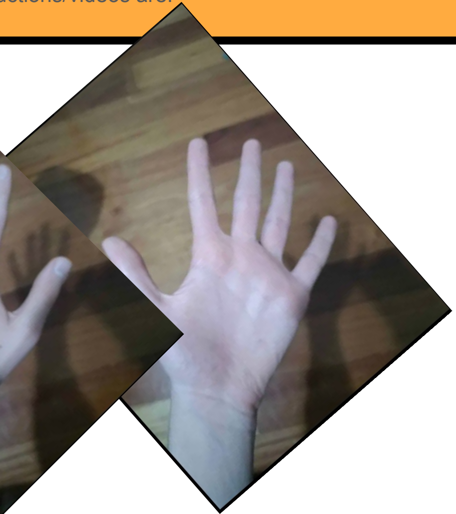


Nuts and bolts:
Bolts are screws with a constant width. They are passed through a hole and on the other side of the pieces of metal, a nut is screw and secured tightly onto the metal

Ergonomics

Overview:

The ergonomics and anthropometrics of my product are important as they dictate how comfortable it is to use, controlling how likely they are to recommend it to others. To do this, I will take measurements from my client and use this anthropometric data to design a produce which is somewhat easy to build, understand and learn from. I need not only take into account o the robot physically feels when building it but also how loud it is when running, how understandable the instructions/videos are.



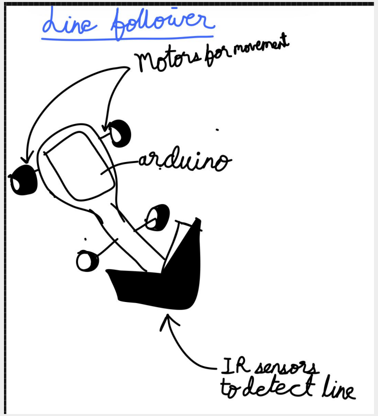
Cognitive ergonomics:
This aspect of the product refers to how the product is perceived. This includes details such as the clarity of the instructions, how the code is presented and how concepts are explained. As an educational kit, this will be one of the most important aspects as without clarity, the box would be a bunch of parts.

Physical Ergonomics:
This is the aspect of the product which the user will physically interact with. Hence, the product must not consist of any sharp corners, the surfaces must be smooth. To keep the surfaces smooth, I will sand the surfaces using gradually increasing grits of sandpaper and use counterboxed nuts and bolts to maintain this smoothness throughout the robot. It should also be easy to transport thus be light. In addition, the robot should also not be extremely loud. This is an important aspect of the robot as often electronic parts can make lots of noise such as magnetic noise (the noise which motors make due to their internals) so to reduce this will make the kit hugely more bearable.

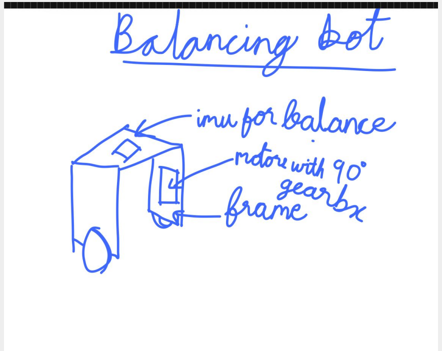
Area	Specific Point
Aesthetics	1.1: My product should look like a unique robot but based on something real **in order to intrigue the audience for which this product is targeted for** 1.2: The colours of the product will be bright and vibrant **to encourage creativity and problem solving.**
Packaging	2.1: The budget will be QAR 150 as **it is a main priority of my product is that it is cheap to reach as many students as possible.** 2.2: I want to price my product at QAR 200 as **it is at price low enough to be affordable but still high enough to make the product high quality.**
Client	3.1 My client is Faris Tarraf as **he already has a keen interest in robotics however, is unable to participate in competitions due to his location.** 3.2: My wider target market is children and teenagers as **they are often building their interests during this time and are often the main target for large competitions..** 3.3: Ideally, my product would be sold on amazon and would work with school to possibly integrate it into the curriculum or as a CCA **as it allows the product to reach the widest target audience.**
Environment	4.1: My product will try to reduce the amount of negative effects it will have on the environment by using much eco-friendly materials as possible as **this will make consumers more likely to buy the product**. 4.2: **Due to the use of electronics,** they product as a whole will not be recyclable. However, **the use of PLA** permits the main body to be recycled and the electrons to be reused..
Size	5.1: When built, the product will be around 20cm x 20cm by 20cm **so it can be held in the hand and can be worked on easily.**
Safety	6.1:The final product will have secure wire connections, protected electronics and double insulation. **This ensures that it is able to get the CE and Kitemark stickers**. Furthermore, it will not have sharp corners and will use less than 12V,
Function	7.1: The kit will be able to educate the user on basic and advanced engineering and robotics concepts such as different types of gears, CPG gaits and Extended Kalman FIlters..
Materials	8.1: The body will be made of PLA and the electronic will be outsourced, most likely using arduino parts.. **As it is cheap and easy to prototype with and the arduino should not quickly degrade.** 8.2: A matte finish will be applied onto the body. **This is to minimise the warping of the body in high temperatures and provides a smooth finish.**

User Requirements	9.1: It must be a programmable, cheap and slightly difficult to assemble as t**o encourage the user to utilise their own critical thinking to solve problems on their own**. 9.2: I will try to keep the wants integrated to a minimum **to reduce costs**.
Maintenance	10.1: The product should not require maintenance as **once its built, the user should have understood all of the concepts. Once this has happened the product has completed its mission.**
Ergonomics	11.1: The corners will be chamfered and the screws will be easily accessible so t**hat everyone, no matter their hand size, can build it..**
Manufacturing	12.1: Most of the product will be 3d printed or laser cut and secured together with Metal nuts and bolts. I may include some wooden panels as well. **This allows for more modularity of the robot and the undoing of steps.**
Product Lifespan	13.1 The product should last about a year at the least. 13.2: Once used, the 3d printed body can be thrown away and the arduino could be used to make other projects.
Packaging	14.1: The product will require packaging as it will be quite delicate with all of the loose electrical components being fragile.

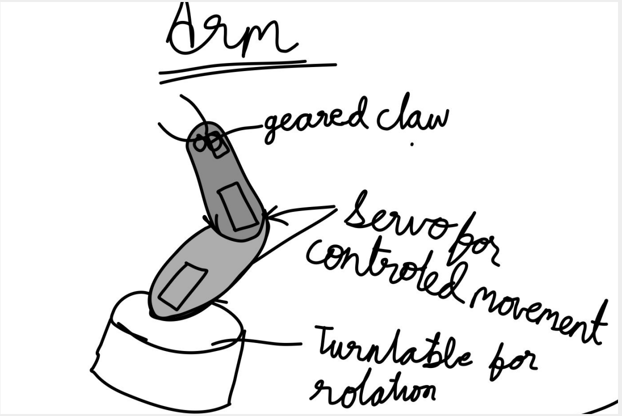
Kit for a line following robot



Kit for a 2-wheel balancing robot

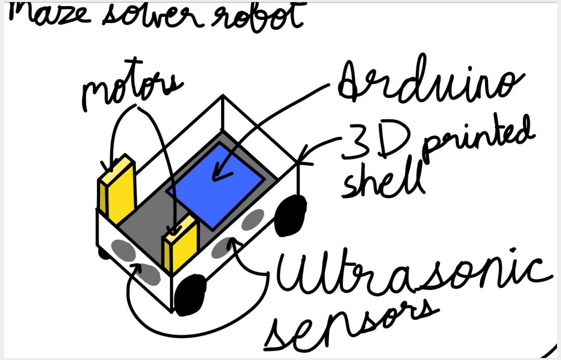


Kit for a robotic arm

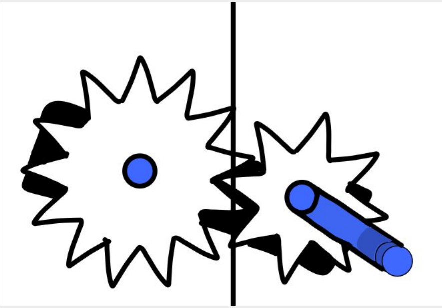


Kit for an Environmental Monitoring robot

Kit for a maze solver robot

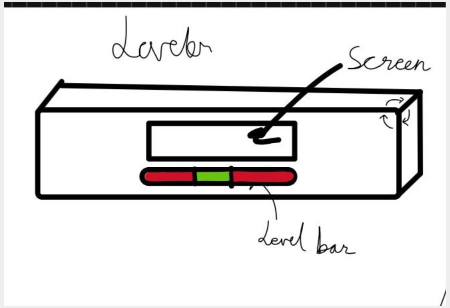


Kit full of separate mechanisms

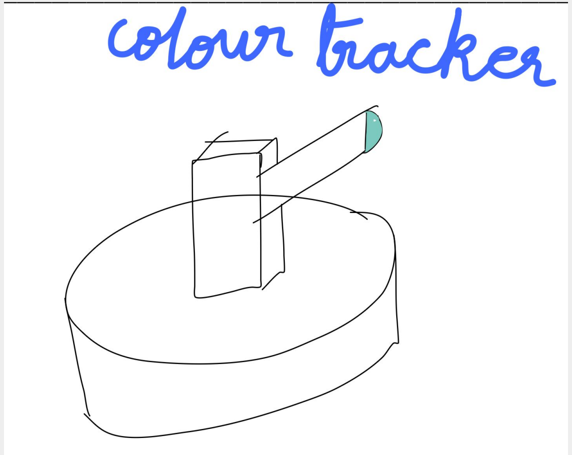


Kit for automatic light switch and curtains

Kit for an electronic leveler

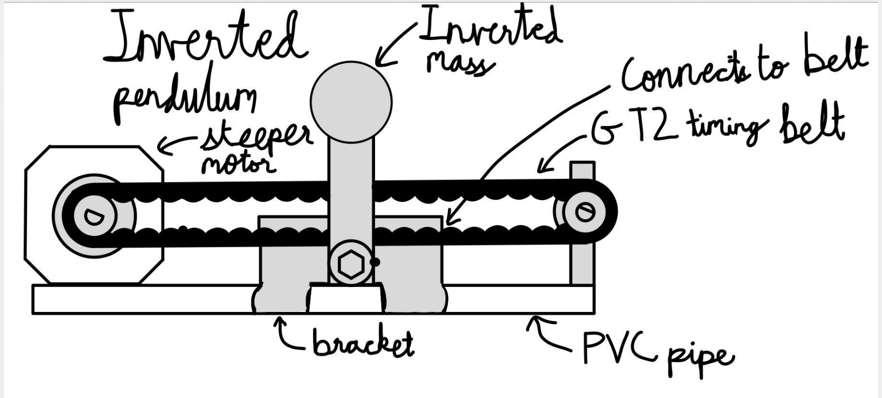


Kit for a colour tracking arm



Kit for a tank drive robot with odometry

Kit for a inverted pendulum



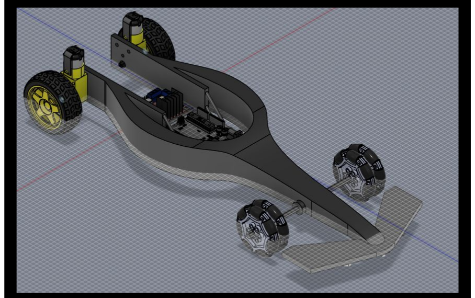
Kit for a bipedal robot

Idea	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Total
1	5	4	4	5	3	5	4	5	3	5	5	5	5	4	62
2	5	4	4	5	5	4	5	3	5	3	4	4	5	5	61
3	4	5	4	3	4	3	3	3	4	5	4	4	4	3	53
4	4	4	3	5	5	4	3	5	4	5	5	4	5	3	59
5	3	5	5	3	5	4	3	4	5	5	4	5	3	5	59
6	5	4	3	4	4	3	3	3	3	4	5	4	3	3	51
7	4	4	4	5	3	3	3	4	5	4	5	5	3	5	57
8	5	4	3	5	4	5	3	5	5	3	5	4	5	3	59
9	4	4	4	4	3	4	4	5	5	3	3	3	4	4	54
10	3	3	4	5	5	5	5	5	4	4	5	3	5	4	60
11	4	4	4	4	5	4	3	4	5	5	5	4	4	5	60
12	4	5	5	3	5	5	4	5	5	4	5	5	3	3	61

Line following robot

The line following robot is a robot which is able to follow a black line against a white background. It uses 2 rear motors to drive the car and 2 sets of amni wheels to allow the car freedom of movement when turning. It uses a set of 3 IR sensors to detect the darkness of the ground below it. It was modeled after and F1 car and students can use it to learn about basic motor usage and how different sensors work.

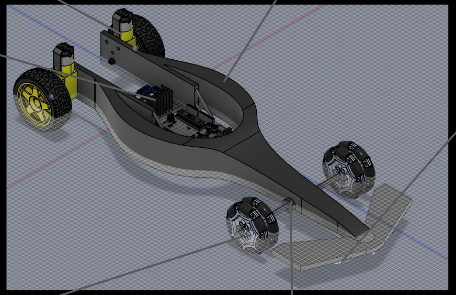
Cardboard prototype of product



TT motors drive the back wheels

The side pods make the car look like an F1 car

The LN298 motor drivers and arduino are situated in the middle to make the physics of the car more similar to F1 cars



IR sensors at the bottom detect changes in the colour of the floor which informs the arduino to turn.

Dual set of omni wheel ensures that at least 1 side-roller is in contact with the ground, preventing increased wheel deterioration

Bushings reduce the friction between the axle and the hole.

Pros and cons

Manufacturing process

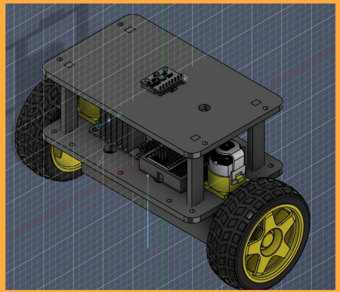
Client feedback

1.1	1.2	2.1	2.2	3.1	3.2	3.3	4.1	4.2	5.1	6.1	7.1	8.1	8.2	9.1	9.2	10.1	11.1	12.1	13.1	13.2	14.1

2 Wheel balancing bot

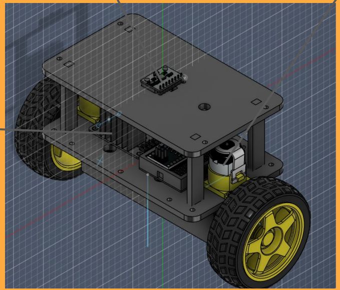
The dual wheel balancing robot is a robot which is able to self align itself upright simply through acceleration of the bottom two wheels. This will be done through the collaboration between an Inertial measurement unit and the arduino. Despise the hardware, the user will have to learn about PIDF and feedback loops to stabilize the machine. This design also allows for more further development of the mechanism as it can be adapted for other uses such as a remote controlled car or an object movement system.

Cardboard prototype of product



MPU 6050 measure the rotational speed which can be converted into an absolute angle.

TT motor drive the body of the robot up by applying lateral motion different to the motion of the CoM



Pros and cons

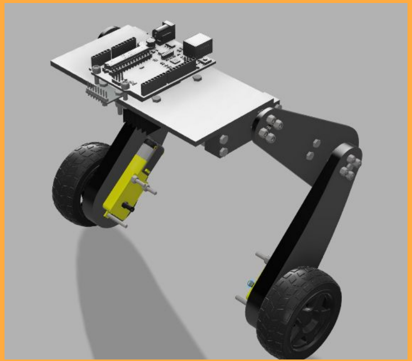
Manufacturing process

Client feedback

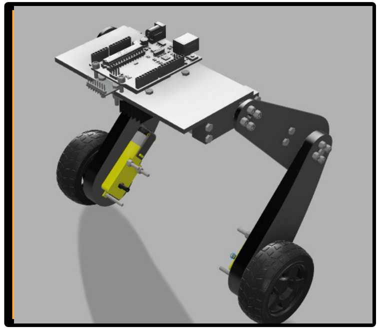
1.1	1.2	2.1	2.2	3.1	3.2	3.3	4.1	4.2	5.1	6.1	7.1	8.1	8.2	9.1	9.2	10.1	11.1	12.1	13.1	13.2	14.1

Bipedal Robot

The concept for this prototype is to act as a racing robot through its freedom of movement. Due to its strong base motors, the robot will be able to sprint along straight but the individual vertical freedom on each leg allows the robot to tilts into corners. Not only this, the robot will be able to ride across terrains without being tipped over. It is through this freedom from which we can teach the users about feedback loops such as PIDF and



Cardboard prototype of product



Pros and cons

Manufacturing process

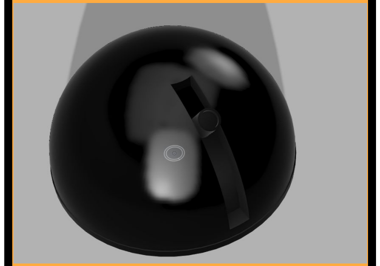
Client feedback

1.1	1.2	2.1	2.2	3.1	3.2	3.3	4.1	4.2	5.1	6.1	7.1	8.1	8.2	9.1	9.2	10.1	11.1	12.1	13.1	13.2	14.1

Colour tracking turret

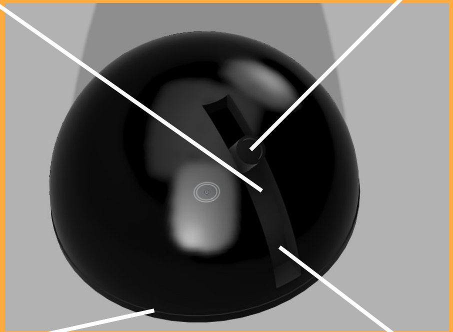
The colour tracking turret is able to autonomously use a camera connected to an arduino to detect a certain colour and follow that colour around using a main servo controlling ϕ and a secondary servo controlling θ in a 3d polar coordinate system.

Cardboard prototype of product



All electronics will be bolted onto the platform to prevent parts from moving when they don't have to.

Camera searches for the largest blob of orange in its view



A Heavy base is used to prevent the motor from moving the base instead of the dome

The whole dome rests on a lazy susan bearing. This allows the whole dome to rotate instead of just the camera

Pros and cons

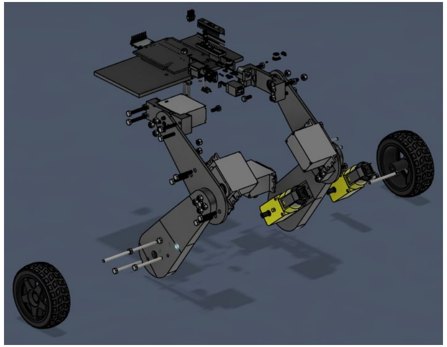
Manufacturing process

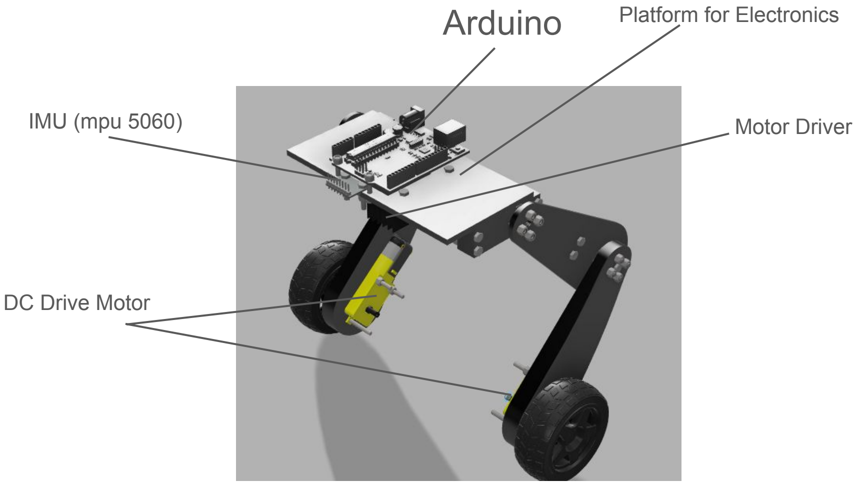
Client feedback

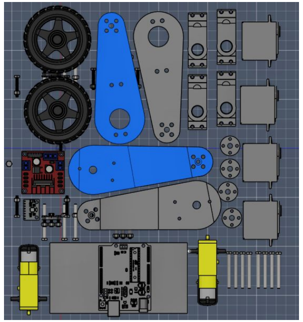
1.1	1.2	2.1	2.2	3.1	3.2	3.3	4.1	4.2	5.1	6.1	7.1	8.1	8.2	9.1	9.2	10.1	11.1	12.1	13.1	13.2	14.1

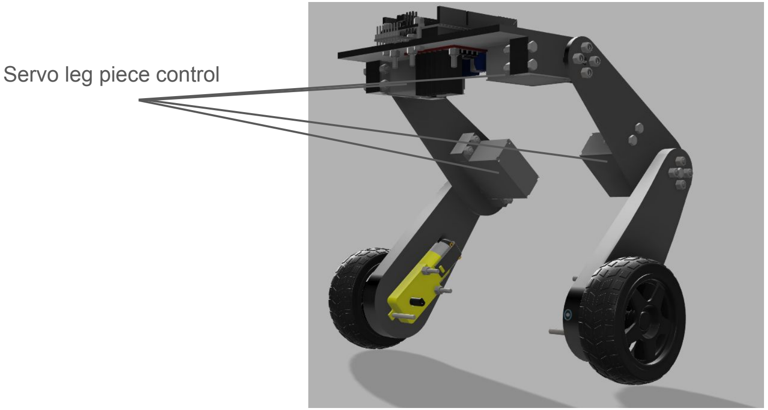
Proposed Solution

The concept for this prototype is to act as a racing robot through its freedom of movement. Due to its strong base motors, the robot will be able to sprint along straight but the individual vertical freedom on each leg allows the robot to tilts into corners. Not only this, the robot will be able to ride across terrains without being tipped over. It is through this freedom from which we can teach the users about feedback loops such as PIDF and the link between maths and engineering through inverse kinematics









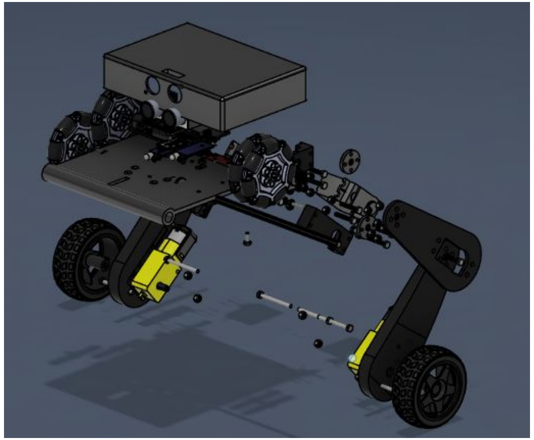
Presentation

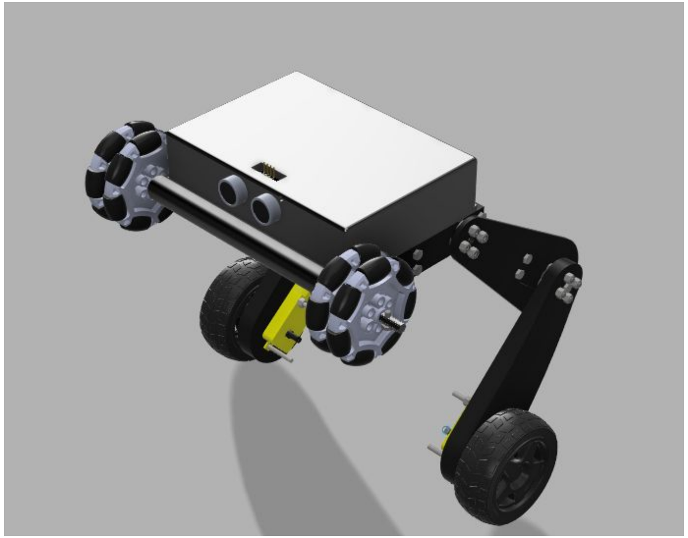
Changes from Design V1

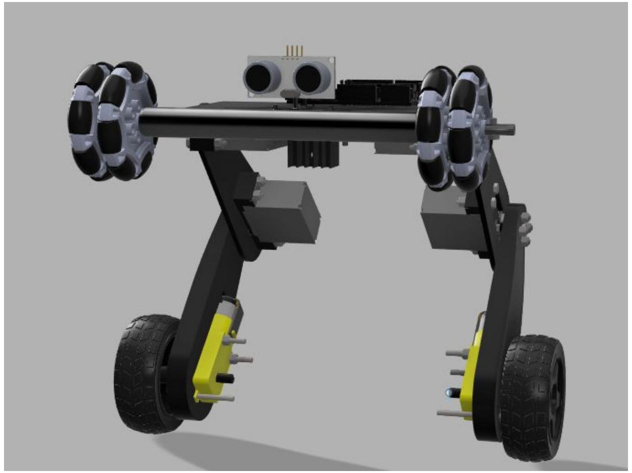
Case around arduino

Remove sharp corners

Rearrange the components

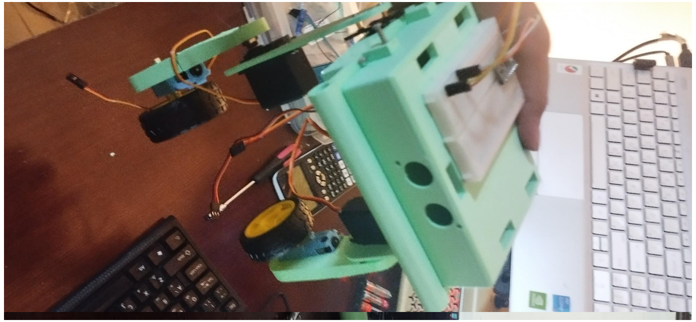






Developed Working Physical Prototype

Group of Photos



Prototype V1.5

Changes made:

Added support wheels for initial testing

Added slot for battery

Developed Prototype V2

Changes made:

Add hole in Lid for USB access

Add holes for Pin access

Allow for 2 batteries

Countersink Screws

Enlarge motor brackets space for tolerance

Further research

PCB Design

Allow for different boards to be mounted

Maybe use a custom board

Deciding between using LQR and PIDF

PIDF vs LQR vs MPC

Conclusion:
Use STM32F4 for main processing and an esp32 for communications